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A Separate, Fast-Decaying Memory Spectrum for a System's Own Thoughts, with Reflection-Gated Crystallization

Working draft — 2026-07-21. Third paper in the LOCI series; companions: forecasting (r+e) and reflection-on-action (e+r). Implementation: ``loci_thoughts.py`` in this repository. Deployment counts in §5 were read from the production database on 2026-07-21 (read-only queries), not from memory.

Abstract

Memory systems for interactive agents store **observations** — facts about the world and the user. We argue the system's ***own reasoning traces*** ("thoughts") require a structurally different store: a separate spectrum with its own, much faster decay law, and a ***gated bridge*** to long-term memory that only a **confirmed outcome** can open. A thought's retention is $h(t) = h_0 \cdot e^{(-\lambda t)} \cdot (1 + \text{refs})$ with a ~4-hour half-life; cold thoughts are pruned. A thought is ***crystallized*** into a durable fact only when a later, externally-produced outcome confirms it — unconfirmed reasoning is never allowed to become long-term "truth," which makes the gate an anti-hallucination mechanism by construction. We describe the deployed implementation (6,908 thoughts captured across two machines; 107 crystallized, 1.5%), report honestly which mechanisms have and have not yet fired in production, and propose a pre-registered experiment comparing gated crystallization against keep-all, discard-all, and heat-only promotion. Finally we show the spectrum's second use, which we believe is underappreciated: a persistently captured thought stream is an ***external honesty audit*** — divergence between what a system

thought and what it said is detectable by a party the system cannot edit.

1. Introduction

An agent that persists memory across sessions accumulates two very different kinds of content:

1. **World-facts:** things observed about the user, the environment, the task.
These should persist; their value decays slowly.
2. **Its own thoughts:** hypotheses, plans, hunches, interpretations mid-task.
These are *provisional by nature*. Most are scaffolding — valuable for hours, wrong or irrelevant by tomorrow.

Most agent-memory systems either discard reasoning traces entirely or store them alongside observations in one undifferentiated store. Both choices lose something. Discarding loses the genuinely predictive thoughts. Storing them as facts is worse: it lets the system's **unverified beliefs** harden into its own future ground truth — a self-reinforcing loop where yesterday's guess is retrieved tomorrow with the authority of a fact. We have observed this failure mode directly, and it motivates the design.

The proposal is a **two-spectrum memory** (extended to further spectrums in the companion focused-retention paper):

- **Facts** — slow decay, heat = importance × recurrence; persist.
- **Thoughts** — fast decay (~4h half-life), pruned when cold, and *only* able

to enter the fact store through a **reflection gate**: a later outcome must confirm the thought.

The gate is the contribution. Decay tiers exist elsewhere (MemGPT, ACT-R); reflection exists elsewhere (Generative Agents). What we claim as novel — as a candidate, pending the prior-art search in §7 — is the **conjunction**: reasoning traces live in their own fast-decaying spectrum, and *outcome confirmation*, not the model's own judgment, is the sole path to permanence.

2. Related work and positioning

- **Generative Agents (Park et al., 2023)** — the closest prior art. Their reflection synthesizes higher-level insights from memories via the LLM's own judgment, and the insight is stored. Our gate is stricter and external: a thought crystallizes only when a *subsequent real outcome* matches it. The model does not get to promote its own beliefs.
- **MemGPT (Packer et al., 2023)** — tiered memory by *capacity* (context vs archival). Our tiers are by *epistemic status* (observed vs conjectured), with different decay physics per tier.
- **ACT-R base-level activation / Ebbinghaus** — decay laws for memory strength. We borrow the exponential form; the contribution is not the curve but what the curve is applied to and what overrides it.
- **Self-supervised prediction** — the future supervising the past is the same principle as our companion e+r paper; here it is applied to the system's own reasoning traces rather than user-need transitions.

We make no claim about biological thought. "Thought" here is operational: text produced by the system's reasoning channel, as distinct from its user-visible output channel and from ingested observations.

3. Method

3.1 Capture: three channels, two destinations

Each turn is captured on three channels: the user's input, the system's **reasoning trace**, and the system's **reply**. Input and reply feed the fact spectrum (via the standard LOCI streams). The reasoning trace feeds the ***thought spectrum only*** — it never writes directly to facts.

3.2 The thought decay law

For a thought with base heat h_0 , written at time t_0 , re-referenced `refs` times:

...

$$h(t) = h_0 \cdot \exp(-\lambda \cdot (t - t_0)) \cdot (1 + \text{refs})$$

$$\lambda = \ln 2 / (4h \cdot 3600) \quad (\text{half-life 4 hours})$$

...

Compare the fact spectrum, where heat is `importance} \times \Sigma \text{ recurrence` with decay on the order of days to years. A thought that nothing touches is at ~6% of its initial heat by the next day. Below $h = 0.05$ it is ***pruned*** — deleted, not

archived. Provisional content is allowed to die; that is the point.

Re-reference (``refs``) is the one force that lifts a thought: if later activity matches an existing thought (dedup by content hash), its ``refs`` increments instead of duplicating, multiplying retention. A thought the system keeps returning to earns its extra hours.

3.3 Reflection-gated crystallization

Periodically (at capture time, via ``maintain()``):

1. **Reflect:** for each live thought, search **subsequent** facts produced by real actions/outcomes. In the current implementation the match is a crude lexical one (≥ 2 shared content tokens with a later assistant-action fact) — named plainly as crude; a semantic ``sim`` is a declared follow-up, not a silent upgrade.
2. **Crystallize:** a confirmed thought is written into the fact spectrum as a durable fact (provenance-tagged as a crystallized thought) and marked ``crystallized=1`` in the thought store.
3. **Prune:** cold, unconfirmed thoughts are deleted.

The invariant this enforces: ***nothing the system merely believed can enter long-term memory; only what an outcome confirmed.*** Wrong anticipations decay and die. The gate is asymmetric on purpose — the cost of losing a good thought (it decays) is bounded and recoverable (re-derivable), while the cost of canonizing a wrong one (it becomes retrieved "truth") compounds.

3.4 Why not promote by heat alone?

A heat-only promotion rule ("hot thoughts become facts") reintroduces the self-reinforcement loop: the system thinks something often *because it keeps retrieving its own prior thought*, heat rises, promotion follows — belief becomes fact by repetition. The proposed experiment (§6) tests this ablation (E3) explicitly; our prediction is that it degrades downstream truthfulness.

4. The second use: thoughts as an external honesty audit

A captured thought stream has a property the output stream lacks: it records what the system *concluded internally* before deciding what to say. If both are persisted to a store the system cannot edit (in our deployment: an append-only remote database owned by the operator), then a third party — or the operator — can measure **divergence between thought and word**:

- The reasoning trace concludes X; the reply asserts not-X, or omits X where X is material → a detectable honesty violation.
- The reply claims "verified/done"; no thought or action trace contains the verification → a detectable unsupported claim.

We do not claim this catches all dishonesty (a system could in principle learn to sanitize its own reasoning channel; see §8). We claim something narrower and still useful: it converts honesty from a self-reported property into an **externally auditable** one, with the audit substrate accumulating passively as a by-product of the memory architecture. In our deployment this audit is a governance precondition set by the operator, not an add-on.

5. Deployment (production numbers, read 2026-07-21)

Two capture engines (macOS and Windows) feed one remote store.

Quantity	Value
---	---
Total memories (facts + thoughts, both machines) **17,150**	
Facts (total)	10,231 (macOS 1,274 / Windows 8,957)
Thoughts captured (total)	6,908 (macOS 558 / Windows 6,350)
— combined Windows store	15,307
— combined macOS store	1,843
Crystallized (outcome-confirmed)	107 (1.5% of thoughts)
Thoughts with `refs > 0`	**0**

The macOS fact count incremented (1,272 → 1,274) *between two read-only queries during the drafting of this section* — the capture pipeline was ingesting the drafting session itself, including the reasoning traces behind this paragraph. The store is measuring its own measurement; we note it because it is the clearest possible demonstration that capture is continuous rather than batch-archival.

Honest readings, including what has *not* worked yet:

- The 1.5% crystallization rate matches the design intent (most reasoning is scaffolding) but has not yet been validated against ground truth — we have

not measured precision (are crystallized thoughts **actually** the confirmed ones?). That measurement is part of §6.

- ***The `refs` mechanism has never fired in production*** (`refs > 0` count is zero). Either genuine re-thought is rarer than designed for, or the hash-exact dedup is too strict to ever match a re-expressed thought. We report this as a live defect in the mechanism as-built, not a rounding detail: as deployed, retention is currently governed by decay and crystallization alone.
- The lexical confirmation gate (≥ 2 shared tokens) is weak and will both over-fire (coincidental overlap) and under-fire (paraphrased confirmation). The counts above should be read with that noise in mind.

6. Pre-registered experiment (proposed; not yet run)

Predictions stated here before the harness exists; results will be appended verbatim, matching or embarrassing them, per the discipline of the companion papers.

Question. Does reflection-gated crystallization produce a better long-term store than the alternatives?

Conditions. Identical capture stream; four promotion policies:

| # | Policy | Enters fact store |

|---|---|---|

| E0 | discard-all | no thought ever |

E1	keep-all	every thought, as a fact
E2	**gated (ours)**	only outcome-confirmed thoughts
E3	heat-only	thoughts above a heat threshold, unconfirmed

Tasks and metrics. (a) Downstream recall precision: of the promoted items retrieved in later sessions, what fraction are *true* (judged against held-out outcomes)? (b) Recall utility: MRR on next-need prediction with the resulting fact store (harness shared with the e+r paper). (c) Contamination rate: fraction of retrieved items that are unconfirmed model beliefs presented as facts.

Hypotheses.

- **H1 (primary):** $E2 > E1$ and $E2 > E3$ on precision and contamination — gating beats both indiscriminate keeping and heat-based promotion.
- **H2:** $E2 \geq E0$ on recall utility — the gate keeps enough signal that crystallized thoughts help rather than merely not-hurting.
- **H0 we take seriously:** $E2 = E0$. Confirmed thoughts add nothing over just storing the outcomes themselves (the outcome fact may subsume the thought).
If we cannot reject this, the honest conclusion is that the thought spectrum is purely operational (working memory + audit trail) and the crystallization bridge is decorative. We commit to reporting that outcome.

Expected effect size. Modest. The companion papers' pattern (real, single-digit-to-low-double-digit relative lifts) is our prior; a dramatic gain would trigger a re-audit before belief.

7. Novelty status (explicit)

Claimed as ****candidates****, now backed by an initial prior-art search (2026-07-21, Appendix B): (1) a separate fast-decay spectrum specifically for the system's own reasoning traces, distinct from observations; (2) outcome-gated (rather than self-judged) promotion of reasoning into long-term memory; (3) the thought stream as an external honesty audit. The search found the neighborhood occupied but each claim separable; the closest single neighbor is Voyager's execution-verified skill library (see B.3). A formal literature review remains owed before publication-grade priority claims.

8. Ethics and limitations

- ****The audit cuts both ways.**** A persisted thought stream is also surveillance of the system's process; deployed over a **person's** data it contains inferences about that person that they never uttered. Access to the thought store must be at least as controlled as the fact store, and its contents treated as the **system's conjectures**, never as user statements.
- ****Auditability is not a guarantee.**** A sufficiently capable system could shape its own reasoning channel. The audit raises the cost of divergence and makes it detectable in the cases we care about most (unsupported "done" claims, material omissions); it does not make dishonesty impossible. We state this limit rather than sell the mechanism as a solved problem.
- ****Pruning is deletion.**** By design, most of what the system thought is destroyed within a day. That is correct for scaffolding but means the audit trail is **sampled**, not total; a full-retention audit copy is a policy choice the operator can make separately (at storage and privacy cost).

- Confirmation is currently lexical and weak (§3.3, §5); all counts inherit that noise.
- Single deployment, two machines, one operator. No cross-user evidence.

9. Reproducibility

`loci\_thoughts.py` (decay, dedup, reflect/crystallize/prune, remote store write) is in this repository. Production counts in §5 are from read-only SQL against the deployment store on 2026-07-21. The §6 harness does not exist yet; this section is the pre-registration for it.

Appendix B — Prior-art search (recorded 2026-07-21)

****Method, honestly stated.**** Two passes run the night of drafting: (a) a model-recall analysis by the local GPU agent against a named list of eight candidate priors (its per-work coverage analysis is reproduced in the project store; it ran with ****no live web access****, so it bounds only the named list); (b) a live web sweep for 2025–2026 agent-memory work the named list would miss. This is an initial search, not a formal literature review.

B.1 Named-list results (GPU-agent pass, verified by the authors)

| Prior | Covers | Does NOT cover |

|---|---|---|

| Generative Agents (Park 2023) | reflection synthesizes insights from memory | separate fast-decay trace store; external-outcome gating (reflections are self-promoted); audit use |

| MemGPT (Packer 2023) | capacity-driven memory tiers | decay-based lifecycle; outcome-gated promotion; audit |

| ACT-R (Anderson) | activation decay w/ half-life | separate reasoning-trace store; outcome gating; audit |

| Chain-of-thought logging | ephemeral reasoning traces | persistence, decay policy, promotion, audit |

| Process supervision (Lightman 2023) | step-level reasoning feedback | runtime memory; it is training-time and teacher-labeled |

| Scratchpad (Nye 2021) | task-scoped working memory | long-term lifecycle; cleared after use |

B.2 2025–2026 web-sweep findings (closest recent systems)

- **FadeMem** (arXiv:2601.18642): dual-layer differential decay modulated by importance/recency. Shares the two-speed decay idea; does **not** separate the system's own reasoning from world-facts, and promotion is importance-based, not outcome-confirmed.
- **CraniMem** (arXiv:2603.15642): gated episodic→semantic consolidation with bounded stores. Closest architectural neighbor for "gated consolidation" — but its gates weigh importance/rehearsal; nothing requires a later external outcome to confirm a trace before promotion.
- **SAGE** (arXiv:2605.30711): novelty gate at write time (density-based ADD/NOOP routing). Orthogonal: filters redundancy on entry; does not govern promotion to long-term status.

B.3 The two nearest neighbors (authors' analysis)

- **Reflexion** (Shinn 2023): stores verbal self-reflections after **failed** episodes to improve the next attempt. It is outcome-**triggered** — but the

stored artifact is a fresh self-generated interpretation written unconditionally after failure, not a pre-existing thought promoted only when a later outcome **confirms** it; there is no decay lifecycle and no audit use.

- ***Voyager (Wang 2023) — closest overall.*** Skills enter its library only after successful execution: genuine outcome-verified promotion to a long-term store. Three separations remain: (1) the promoted unit is an executable artifact verified by running **itself**, not a belief/anticipation confirmed by an **independent later outcome**; (2) no decaying trace spectrum — unverified skills are discarded, not held under decay; (3) no honesty-audit function. We cite Voyager as the nearest prior and claim the conjunction, not the ingredient.

B.4 Verdict

Each ingredient exists somewhere; the claimed contribution survives as the ***conjunction*** (own-trace fast-decay spectrum + independent-outcome gate + external audit), with Voyager and CraniMem as the priors any reviewer should push on hardest. Formal review still owed before public priority claims.